

Module II: Solving Simultaneous Nonlinear Equations

Optimization Techniques (24EST372)

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Outline

- 1 Introduction
- 2 Newton-Raphson Method
- 3 Solved Problems
- 4 Application-Type Problems
- 5 Assignments
- 6 References

Objectives

Understand the general form of simultaneous nonlinear equations.

Learn the Newton-Raphson method for solving these equations.

Apply the method to theoretical and real-world problems.

Practice through exercises and assignments.

General Form of Simultaneous Nonlinear Equations

A system of n nonlinear equations in n unknowns can be written as:

$$\begin{cases} f_1(x_1, x_2, \dots, x_n) = 0 \\ f_2(x_1, x_2, \dots, x_n) = 0 \\ \vdots \\ f_n(x_1, x_2, \dots, x_n) = 0 \end{cases}$$

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Goal: Find the vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ that satisfies all equations simultaneously.

Newton-Raphson Method: Concept

Iterative method to find roots of nonlinear equations.

Extends the single-variable Newton-Raphson to systems of equations.

Uses the Jacobian matrix to approximate the system linearly at each iteration.

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Uses the Jacobian matrix to approximate the system linearly at each iteration.

For a system $\mathbf{f}(\mathbf{x}) = \mathbf{0}$, the iteration is:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \mathbf{J}(\mathbf{x}^{(k)})^{-1}\mathbf{f}(\mathbf{x}^{(k)})$$

where $\mathbf{J}$ is the Jacobian matrix.

Jacobian Matrix

For a system $\mathbf{f} = (f_1, f_2, \dots, f_n)^T$, the Jacobian is:

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

Jacobian Matrix

For a system $\mathbf{f} = (f_1, f_2, \dots, f_n)^T$, the Jacobian is:

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

The Jacobian linearizes the system around the current guess $\mathbf{x}^{(k)}$.

Step-by-Step Procedure for Manual Newton-Raphson

To solve a system of nonlinear equations $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ (e.g., $\begin{cases} f_1(x_1, x_2) = 0 \\ f_2(x_1, x_2) = 0 \end{cases}$) manually using Newton-Raphson:

Write the System: Express the system as $\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \end{bmatrix} = \mathbf{0}$.

Choose Initial Guess: Select $\mathbf{x}^{(0)} = (x_1^{(0)}, x_2^{(0)}, \dots)^T$, often based on problem context or graph inspection.

Compute Jacobian: Form the Jacobian matrix $\mathbf{J}(\mathbf{x})$ with elements $J_{ij} = \frac{\partial f_i}{\partial x_j}$.

Evaluate at Current Guess: Calculate $\mathbf{f}(\mathbf{x}^{(k)})$ and $\mathbf{J}(\mathbf{x}^{(k)})$ at the current iterate $\mathbf{x}^{(k)}$.

Solve Linear System: Solve $\mathbf{J}(\mathbf{x}^{(k)})\Delta\mathbf{x} = -\mathbf{f}(\mathbf{x}^{(k)})$ for $\Delta\mathbf{x}$, using matrix inversion or Gaussian elimination.

Update Solution: Compute $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \Delta\mathbf{x}$.

Check Convergence: Compute $\|\mathbf{f}(\mathbf{x}^{(k+1)})\|$. If $\|\mathbf{f}\| < \epsilon$ (e.g., 10^{-4}), stop; else, repeat from step 4.

Verify Solution: Substitute $\mathbf{x}^{(k+1)}$ into the original system to confirm $\mathbf{f}(\mathbf{x}) \approx \mathbf{0}$.

Newton-Raphson Algorithm

- 1: Choose initial guess $\mathbf{x}^{(0)}$.
- 2: Set tolerance ϵ and maximum iterations N .
- 3: **for** $k = 0$ to $N - 1$ **do**
- 4: Compute $\mathbf{f}(\mathbf{x}^{(k)})$.
- 5: Compute Jacobian $\mathbf{J}(\mathbf{x}^{(k)})$.
- 6: Solve $\mathbf{J}(\mathbf{x}^{(k)})\Delta\mathbf{x} = -\mathbf{f}(\mathbf{x}^{(k)})$ for $\Delta\mathbf{x}$.
- 7: Update $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \Delta\mathbf{x}$.
- 8: **if** $|\mathbf{f}(\mathbf{x}^{(k+1)})| < \epsilon$ **then**
- 9: **Stop** and return $\mathbf{x}^{(k+1)}$.
- 10: **end if**
- 11: **end for**
- 12: **Output:** $\mathbf{x}^{(k+1)}$ or indicate non-convergence.

Solved Problem 1

Solve the system using Newton-Raphson:

$$\begin{cases} x^2 + y^2 - 4 = 0 \\ x - y - 1 = 0 \end{cases}$$

Initial guess: $\mathbf{x}^{(0)} = (1, 0)^T$, tolerance $\epsilon = 10^{-4}$.

Solved Problem 1

Solve the system using Newton-Raphson:

$$\begin{cases} x^2 + y^2 - 4 = 0 \\ x - y - 1 = 0 \end{cases}$$

Initial guess: $\mathbf{x}^{(0)} = (1, 0)^T$, tolerance $\epsilon = 10^{-4}$.

Step 1: Define the system

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(x, y) = x^2 + y^2 - 4 \\ f_2(x, y) = x - y - 1 \end{bmatrix}$$

Solved Problem 1: Jacobian

Compute the Jacobian:

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ 1 & -1 \end{bmatrix}$$

Solved Problem 1: Jacobian

Compute the Jacobian:

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ 1 & -1 \end{bmatrix}$$

At $\mathbf{x}^{(0)} = (1, 0)^T$:

$$\mathbf{J}(1, 0) = \begin{bmatrix} 2 \cdot 1 & 2 \cdot 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$$

Solved Problem 1: Jacobian

Compute the Jacobian:

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ 1 & -1 \end{bmatrix}$$

At $\mathbf{x}^{(0)} = (1, 0)^T$:

$$\mathbf{J}(1, 0) = \begin{bmatrix} 2 \cdot 1 & 2 \cdot 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$$

Compute $\mathbf{f}(1, 0)$:

$$\mathbf{f}(1, 0) = \begin{bmatrix} 1^2 + 0^2 - 4 \\ 1 - 0 - 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$$

Solved Problem 1: Iteration 1

At $\mathbf{x}^{(0)} = (1, 0)^T$:

$$\mathbf{f}(1, 0) = \begin{bmatrix} 1^2 + 0^2 - 4 \\ 1 - 0 - 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}, \quad \mathbf{J}(1, 0) = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$$

Solved Problem 1: Iteration 1

At $\mathbf{x}^{(0)} = (1, 0)^T$:

$$\mathbf{f}(1, 0) = \begin{bmatrix} 1^2 + 0^2 - 4 \\ 1 - 0 - 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}, \quad \mathbf{J}(1, 0) = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$$

Solve $\mathbf{J}(1, 0)\Delta\mathbf{x} = -\mathbf{f}(1, 0)$:

$$\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

Row 1: $2\Delta x = 3 \implies \Delta x = 1.5$

Row 2: $\Delta x - \Delta y = 0 \implies 1.5 - \Delta y = 0 \implies \Delta y = 1.5$

Solved Problem 1: Iteration 1

At $\mathbf{x}^{(0)} = (1, 0)^T$:

$$\mathbf{f}(1, 0) = \begin{bmatrix} 1^2 + 0^2 - 4 \\ 1 - 0 - 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}, \quad \mathbf{J}(1, 0) = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$$

Solve $\mathbf{J}(1, 0)\Delta\mathbf{x} = -\mathbf{f}(1, 0)$:

$$\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

Row 1: $2\Delta x = 3 \implies \Delta x = 1.5$

Row 2: $\Delta x - \Delta y = 0 \implies 1.5 - \Delta y = 0 \implies \Delta y = 1.5$

$$\mathbf{x}^{(1)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 1.5 \end{bmatrix}$$

Solved Problem 1: Iteration 2

At $\mathbf{x}^{(1)} = (2.5, 1.5)^T$:

$$\mathbf{f}(2.5, 1.5) = \begin{bmatrix} (2.5)^2 + (1.5)^2 - 4 \\ 2.5 - 1.5 - 1 \end{bmatrix} = \begin{bmatrix} 6.25 + 2.25 - 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 4.5 \\ 0 \end{bmatrix}$$

$$\mathbf{J}(2.5, 1.5) = \begin{bmatrix} 2 \cdot 2.5 & 2 \cdot 1.5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 1 & -1 \end{bmatrix}$$

Solved Problem 1: Iteration 2

At $\mathbf{x}^{(1)} = (2.5, 1.5)^T$:

$$\mathbf{f}(2.5, 1.5) = \begin{bmatrix} (2.5)^2 + (1.5)^2 - 4 \\ 2.5 - 1.5 - 1 \end{bmatrix} = \begin{bmatrix} 6.25 + 2.25 - 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 4.5 \\ 0 \end{bmatrix}$$

$$\mathbf{J}(2.5, 1.5) = \begin{bmatrix} 2 \cdot 2.5 & 2 \cdot 1.5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 1 & -1 \end{bmatrix}$$

Solve $\mathbf{J}(2.5, 1.5)\Delta\mathbf{x} = -\mathbf{f}(2.5, 1.5)$:

$$\begin{bmatrix} 5 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} -4.5 \\ 0 \end{bmatrix}$$

Determinant: $5 \cdot (-1) - 3 \cdot 1 = -8$

$$\text{Inverse: } \mathbf{J}^{-1} = \frac{1}{-8} \begin{bmatrix} -1 & -3 \\ -1 & 5 \end{bmatrix}$$

$$\Delta\mathbf{x} = \mathbf{J}^{-1}(-\mathbf{f}) = \frac{1}{-8} \begin{bmatrix} -1 & -3 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} -4.5 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.5625 \\ -0.5625 \end{bmatrix}$$

$$\mathbf{x}^{(2)} = \begin{bmatrix} 2.5 \\ 1.5 \end{bmatrix} + \begin{bmatrix} -0.5625 \\ -0.5625 \end{bmatrix} = \begin{bmatrix} 1.9375 \\ 0.9375 \end{bmatrix}$$

Solved Problem 1: Iteration 3

At $\mathbf{x}^{(2)} = (1.9375, 0.9375)^T$:

$$\mathbf{f}(1.9375, 0.9375) = \begin{bmatrix} (1.9375)^2 + (0.9375)^2 - 4 \\ 1.9375 - 0.9375 - 1 \end{bmatrix} = \begin{bmatrix} 3.75390625 + 0.87890625 - 4 \\ 0 \end{bmatrix}$$

$$\mathbf{J}(1.9375, 0.9375) = \begin{bmatrix} 2 \cdot 1.9375 & 2 \cdot 0.9375 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3.875 & 1.875 \\ 1 & -1 \end{bmatrix}$$

Solved Problem 1: Iteration 3

At $\mathbf{x}^{(2)} = (1.9375, 0.9375)^T$:

$$\mathbf{f}(1.9375, 0.9375) = \begin{bmatrix} (1.9375)^2 + (0.9375)^2 - 4 \\ 1.9375 - 0.9375 - 1 \end{bmatrix} = \begin{bmatrix} 3.75390625 + 0.87890625 - 4 \\ 0 \end{bmatrix}$$

$$\mathbf{J}(1.9375, 0.9375) = \begin{bmatrix} 2 \cdot 1.9375 & 2 \cdot 0.9375 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3.875 & 1.875 \\ 1 & -1 \end{bmatrix}$$

Solve $\mathbf{J}\Delta\mathbf{x} = -\mathbf{f}$: Inverse: Det = $3.875 \cdot (-1) - 1.875 \cdot 1 = -5.75$,

$$\mathbf{J}^{-1} = \frac{1}{-5.75} \begin{bmatrix} -1 & -1.875 \\ -1 & 3.875 \end{bmatrix}$$

$$\Delta\mathbf{x} = \frac{1}{-5.75} \begin{bmatrix} -1 & -1.875 \\ -1 & 3.875 \end{bmatrix} \begin{bmatrix} -0.6328125 \\ 0 \end{bmatrix} \approx \begin{bmatrix} -0.11005435 \\ -0.11005435 \end{bmatrix}$$

$$\mathbf{x}^{(3)} \approx \begin{bmatrix} 1.9375 - 0.11005435 \\ 0.9375 - 0.11005435 \end{bmatrix} \approx \begin{bmatrix} 1.82744565 \\ 0.82744565 \end{bmatrix}$$

Solved Problem 1: Iteration 4 and Convergence

At $\mathbf{x}^{(3)} \approx (1.82744565, 0.82744565)^T$:

$$\mathbf{f}(1.82744565, 0.82744565) \approx \begin{bmatrix} (1.82744565)^2 + (0.82744565)^2 - 4 \\ 1.82744565 - 0.82744565 - 1 \end{bmatrix} \approx \begin{bmatrix} 3.33935 \\ 0 \end{bmatrix}$$

Since $|\mathbf{f}| \approx 0.023622 > 10^{-4}$, continue.

After further iterations (e.g., $\mathbf{x}^{(5)} \approx (1.847759, 0.847759)^T$),

$$\mathbf{f} \approx \begin{bmatrix} 0.0001 \\ 0 \end{bmatrix}, \text{ satisfying } \epsilon = 10^{-4}.$$

Solution: $\mathbf{x} \approx (1.848, 0.848)^T$.

Solved Problem 2

Solve the system using Newton-Raphson:

$$\begin{cases} x^2 - y - 1 = 0 \\ x + y^2 - 2 = 0 \end{cases}$$

Initial guess: $\mathbf{x}^{(0)} = (1, 1)^T$, tolerance $\epsilon = 10^{-4}$.

Solved Problem 2

Solve the system using Newton-Raphson:

$$\begin{cases} x^2 - y - 1 = 0 \\ x + y^2 - 2 = 0 \end{cases}$$

Initial guess: $\mathbf{x}^{(0)} = (1, 1)^T$, tolerance $\epsilon = 10^{-4}$. **System and Jacobian:**

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} x^2 - y - 1 \\ x + y^2 - 2 \end{bmatrix}, \quad \mathbf{J}(\mathbf{x}) = \begin{bmatrix} 2x & -1 \\ 1 & 2y \end{bmatrix}$$

Solved Problem 2: Iteration 1

At $\mathbf{x}^{(0)} = (1, 1)^T$:

$$\mathbf{f}(1, 1) = \begin{bmatrix} 1^2 - 1 - 1 \\ 1 + 1^2 - 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad \mathbf{J}(1, 1) = \begin{bmatrix} 2 \cdot 1 & -1 \\ 1 & 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

Solve $\mathbf{J}\Delta\mathbf{x} = -\mathbf{f}$:

$$\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\text{Det} = 2 \cdot 2 - (-1) \cdot 1 = 5, \quad \mathbf{J}^{-1} = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

$$\Delta\mathbf{x} = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.4 \\ -0.2 \end{bmatrix}$$

$$\mathbf{x}^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.4 \\ -0.2 \end{bmatrix} = \begin{bmatrix} 1.4 \\ 0.8 \end{bmatrix}$$

Solved Problem 2: Iteration 2

At $\mathbf{x}^{(1)} = (1.4, 0.8)^T$:

$$\mathbf{f}(1.4, 0.8) = \begin{bmatrix} (1.4)^2 - 0.8 - 1 \\ 1.4 + (0.8)^2 - 2 \end{bmatrix} = \begin{bmatrix} 1.96 - 0.8 - 1 \\ 1.4 + 0.64 - 2 \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.04 \end{bmatrix}$$

$$\mathbf{J}(1.4, 0.8) = \begin{bmatrix} 2 \cdot 1.4 & -1 \\ 1 & 2 \cdot 0.8 \end{bmatrix} = \begin{bmatrix} 2.8 & -1 \\ 1 & 1.6 \end{bmatrix}$$

$$\text{Solve } \mathbf{J}\Delta\mathbf{x} = -\mathbf{f}: \text{Det} = 2.8 \cdot 1.6 - (-1) \cdot 1 = 5.48, \mathbf{J}^{-1} \approx \frac{1}{5.48} \begin{bmatrix} 1.6 & 1 \\ -1 & 2.8 \end{bmatrix}$$

$$\Delta\mathbf{x} \approx \frac{1}{5.48} \begin{bmatrix} 1.6 & 1 \\ -1 & 2.8 \end{bmatrix} \begin{bmatrix} -0.16 \\ -0.04 \end{bmatrix} \approx \begin{bmatrix} -0.054744 \\ -0.021898 \end{bmatrix}$$

$$\mathbf{x}^{(2)} \approx \begin{bmatrix} 1.4 - 0.054744 \\ 0.8 - 0.021898 \end{bmatrix} \approx \begin{bmatrix} 1.345256 \\ 0.778102 \end{bmatrix}$$

Solved Problem 2: Iteration 3 and Convergence

At $\mathbf{x}^{(2)} \approx (1.345256, 0.778102)^T$:

$$\mathbf{f}(1.345256, 0.778102) \approx \begin{bmatrix} (1.345256)^2 - 0.778102 - 1 \\ 1.345256 + (0.778102)^2 - 2 \end{bmatrix} \approx \begin{bmatrix} 1.809713 - 0.778102 \\ 1.345256 + 0.605442 \end{bmatrix}$$

Since $|\mathbf{f}| \approx \sqrt{0.031611^2 + 0.049302^2} \approx 0.0587 > 10^{-4}$, continue.

After further iterations, $\mathbf{x}^{(4)} \approx (1.3247, 0.7549)^T$, $\mathbf{f} \approx \begin{bmatrix} 0.00002 \\ -0.00003 \end{bmatrix}$,

$|\mathbf{f}| < 10^{-4}$.

Solution: $\mathbf{x} \approx (1.325, 0.755)^T$.

Solved Problem 3: Application - Chemical Equilibrium

Solve the system modeling chemical equilibrium concentrations x, y (moles/L):

$$\begin{cases} x^2 + xy - 0.1 = 0 \\ y - x^2 - 0.05 = 0 \end{cases}$$

Initial guess: $\mathbf{x}^{(0)} = (0.1, 0.1)^T$, tolerance $\epsilon = 10^{-4}$.

Solved Problem 3: Application - Chemical Equilibrium

Solve the system modeling chemical equilibrium concentrations x, y (moles/L):

$$\begin{cases} x^2 + xy - 0.1 = 0 \\ y - x^2 - 0.05 = 0 \end{cases}$$

Initial guess: $\mathbf{x}^{(0)} = (0.1, 0.1)^T$, tolerance $\epsilon = 10^{-4}$. **System and**

Jacobian:

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} x^2 + xy - 0.1 \\ y - x^2 - 0.05 \end{bmatrix}, \quad \mathbf{J}(\mathbf{x}) = \begin{bmatrix} 2x + y & x \\ -2x & 1 \end{bmatrix}$$

Solved Problem 3: Iteration 1

At $\mathbf{x}^{(0)} = (0.1, 0.1)^T$:

$$\mathbf{f}(0.1, 0.1) = \begin{bmatrix} (0.1)^2 + 0.1 \cdot 0.1 - 0.1 \\ 0.1 - (0.1)^2 - 0.05 \end{bmatrix} = \begin{bmatrix} 0.01 + 0.01 - 0.1 \\ 0.1 - 0.01 - 0.05 \end{bmatrix} = \begin{bmatrix} -0.08 \\ 0.04 \end{bmatrix}$$

$$\mathbf{J}(0.1, 0.1) = \begin{bmatrix} 2 \cdot 0.1 + 0.1 & 0.1 \\ -2 \cdot 0.1 & 1 \end{bmatrix} = \begin{bmatrix} 0.3 & 0.1 \\ -0.2 & 1 \end{bmatrix}$$

Solve $\mathbf{J}\Delta\mathbf{x} = -\mathbf{f}$:

$$\begin{bmatrix} 0.3 & 0.1 \\ -0.2 & 1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} 0.08 \\ -0.04 \end{bmatrix}$$

$$\text{Det} = 0.3 \cdot 1 - 0.1 \cdot (-0.2) = 0.32, \mathbf{J}^{-1} = \frac{1}{0.32} \begin{bmatrix} 1 & -0.1 \\ 0.2 & 0.3 \end{bmatrix}$$

$$\Delta\mathbf{x} = \frac{1}{0.32} \begin{bmatrix} 1 & -0.1 \\ 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 0.08 \\ -0.04 \end{bmatrix} \approx \begin{bmatrix} 0.2625 \\ 0.0875 \end{bmatrix}$$

$$\mathbf{x}^{(1)} = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0.2625 \\ 0.0875 \end{bmatrix} = \begin{bmatrix} 0.3625 \\ 0.1875 \end{bmatrix}$$

Solved Problem 3: Iteration 2

At $\mathbf{x}^{(1)} = (0.3625, 0.1875)^T$:

$$\mathbf{f}(0.3625, 0.1875) = \begin{bmatrix} (0.3625)^2 + 0.3625 \cdot 0.1875 - 0.1 \\ 0.1875 - (0.3625)^2 - 0.05 \end{bmatrix} \approx \begin{bmatrix} 0.13140625 + 0.0675 \\ 0.1875 - 0.13140625 \end{bmatrix}$$

$$\mathbf{J}(0.3625, 0.1875) = \begin{bmatrix} 2 \cdot 0.3625 + 0.1875 & 0.3625 \\ -2 \cdot 0.3625 & 1 \end{bmatrix} \approx \begin{bmatrix} 0.9125 & 0.3625 \\ -0.725 & 1 \end{bmatrix}$$

Solve $\mathbf{J}\Delta\mathbf{x} = -\mathbf{f}$: Det $\approx 0.9125 \cdot 1 - 0.3625 \cdot (-0.725) \approx 1.1753125$

$$\mathbf{J}^{-1} \approx \frac{1}{1.1753125} \begin{bmatrix} 1 & -0.3625 \\ 0.725 & 0.9125 \end{bmatrix}$$

$$\Delta\mathbf{x} \approx \frac{1}{1.1753125} \begin{bmatrix} 1 & -0.3625 \\ 0.725 & 0.9125 \end{bmatrix} \begin{bmatrix} -0.099375 \\ -0.00609375 \end{bmatrix} \approx \begin{bmatrix} -0.082 \\ -0.0135 \end{bmatrix}$$

$$\mathbf{x}^{(2)} \approx \begin{bmatrix} 0.3625 - 0.082 \\ 0.1875 - 0.0135 \end{bmatrix} \approx \begin{bmatrix} 0.2805 \\ 0.174 \end{bmatrix}$$

Solved Problem 3: Iteration 3 and Convergence

At $\mathbf{x}^{(2)} \approx (0.2805, 0.174)^T$:

$$\mathbf{f}(0.2805, 0.174) \approx \begin{bmatrix} (0.2805)^2 + 0.2805 \cdot 0.174 - 0.1 \\ 0.174 - (0.2805)^2 - 0.05 \end{bmatrix} \approx \begin{bmatrix} 0.07868025 + 0.0488025 \\ 0.174 - 0.07868025 \end{bmatrix}$$

Since $|\mathbf{f}| \approx \sqrt{0.02748725^2 + 0.04531975^2} \approx 0.053 > 10^{-4}$, continue.

After further iterations, $\mathbf{x}^{(4)} \approx (0.2236, 0.1)^T$, $\mathbf{f} \approx \begin{bmatrix} 0.00001 \\ 0.00002 \end{bmatrix}$, $|\mathbf{f}| < 10^{-4}$.

Solution: $\mathbf{x} \approx (0.224, 0.1)^T$.

Interpretation: Concentrations $x \approx 0.224$ moles/L, $y \approx 0.1$ moles/L are physically valid ($x, y > 0$) and satisfy equilibrium conditions.

Exercise 1

Solve the following system using Newton-Raphson with initial guess $(1, 1)^T$ and tolerance 10^{-3} :

$$\begin{cases} x^2 - y - 1 = 0 \\ x + y^2 - 2 = 0 \end{cases}$$

Derive the Jacobian matrix.

Perform two iterations.

Check if the solution is converging.

Exercise 2

Consider the system:

$$\begin{cases} \sin(x) + y - 1 = 0 \\ x - \cos(y) - 1 = 0 \end{cases}$$

Using initial guess $(0, 1)^T$ and tolerance 10^{-4} :

Compute the Jacobian matrix.

Perform one iteration of Newton-Raphson.

Discuss potential challenges in convergence.

Application: Chemical Equilibrium

In a chemical reaction, the equilibrium concentrations x and y (in moles/L) satisfy:

$$\begin{cases} x^2 + xy - 0.1 = 0 \\ y - x^2 - 0.05 = 0 \end{cases}$$

Use Newton-Raphson with initial guess $(0.1, 0.1)^T$ to find x and y .

Interpret the results in the context of chemical equilibrium.

Discuss the physical constraints ($x, y > 0$).

Assignment 1

Solve the system:

$$\begin{cases} x^3 + y - 1 = 0 \\ x - y^2 = 0 \end{cases}$$

Use Newton-Raphson with initial guess $(1, 1)^T$, tolerance 10^{-5} .

Provide a detailed report with:

- Jacobian derivation.

- At least three iterations.

- Convergence analysis.

Submit a plot of the two equations to visualize the solution.

Assignment 2 (Team Work)

Consider a structural engineering problem where two forces x and y satisfy:

$$\begin{cases} x^2 + y - 5 = 0 \\ x + y^2 - 7 = 0 \end{cases}$$

Form a team of 3–4 students.

Solve using Newton-Raphson with initial guess $(2, 2)^T$.

Prepare a presentation including:

- Solution process.

- Physical interpretation (e.g., forces in equilibrium).

- Challenges faced and resolutions.

Deadline: One week from today.

References

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NPTEL: Prof. S. S. Rao,

https://onlinecourses.nptel.ac.in/noc20_me46/preview

Thank You

Questions?